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Parallel HPC - HW2 - Threads
[Link to Repo]

Assignment1

```
milenat@ubuntu-aua-os:~/hpc/homework/hpc_homework/hw2-threads$ gcc assignment1.c -o a1 -pthread
milenat@ubuntu-aua-os:~/hpc/homework/hpc_homework/hw2-threads$ ./a1
Thread 1 is running
Thread 2 is running
Thread 3 is running
All threads are complete
```

Assignment2

```
milenat@ubuntu-aua-os:~/hpc/homework/hpc_homework/hw2-threads$ gcc assignment2.c -o a2 -pthread
milenat@ubuntu-aua-os:~/hpc/homework/hpc_homework/hw2-threads$ ./a2
Sequential sum: 543666294849
Parallel sum (local sums): 543666294849
Parallel sum (shared+mutex): 543666294849
Sequential time: 0.000003 sec
Parallel time (local sums): 0.000453 sec
Parallel time (shared+mutex): 0.000232 sec
```

Assignment3

```
milenat@ubuntu-aua-os:~/hpc/homework/hpc_homework/hw2-threads$ gcc assignment3.c -o a3 -pthread
milenat@ubuntu-aua-os:~/hpc/homework/hpc_homework/hw2-threads$ ./a3
Sequential max: 2147483512
Parallel max (4 threads): 2147483512
Sequential time: 0.063712 sec
Parallel time:   0.075295 sec
```

Assignment4

```
milenat@ubuntu-aua-os:~/hpc/homework/hpc_homework/hw2-threads$ gcc assignment4.c -o a4 -pthread
milenat@ubuntu-aua-os:~/hpc/homework/hpc_homework/hw2-threads$ ./a4
Sequential primes: 1270607
Parallel primes (4 threads): 1270607
Sequential time: 4.547954 sec
Parallel time:   4.497379 sec
```

Assignment5

```
milenat@ubuntu-aua-os:~/hpc/homework/hpc_homework/hw2-threads$ gcc assignment5.c -o a5 -pthread
milenat@ubuntu-aua-os:~/hpc/homework/hpc_homework/hw2-threads$ ./a5
thread 0 ran on cpu 1
thread 6 ran on cpu 0
thread 1 ran on cpu 0
thread 3 ran on cpu 1
thread 7 ran on cpu 1
thread 4 ran on cpu 1
thread 2 ran on cpu 0
thread 5 ran on cpu 1
milenat@ubuntu-aua-os:~/hpc/homework/hpc_homework/hw2-threads$ taskset -c 0,1,2 ./a5
thread 0 ran on cpu 0
thread 1 ran on cpu 1
thread 2 ran on cpu 1
thread 4 ran on cpu 0
thread 6 ran on cpu 1
thread 5 ran on cpu 0
thread 3 ran on cpu 0
thread 7 ran on cpu 1
```